

ARTICLE

Evidence of negative density-dependent dispersal in an invasive forest pest

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Genome Canada, Grant/Award Number: 18202; Genome Alberta, Grant/Award Number: L20TF

Handling Editor: Samniqueka J. Halsey**Abstract**

Understanding the mountain pine beetle's dispersal patterns is critical for evaluating its threat to Canada's boreal forests. It is generally believed that higher beetle densities lead to increased long-distance dispersal. One possible explanation is that beetle aggregation pheromones become repellent at high concentrations, causing beetles to fly above the forest canopy where wind currents may carry them great distances. Using helicopter surveys of infested trees from 2010 to 2020 in Alberta, we modeled the spatial relationship between successive annual infestations as a proxy for beetle dispersal. Instead of the expected positive association, we observe a weak negative association at all spatial scales: At higher infestation densities, new infestations appear closer to previous years' infestations. A possible explanation is that at low population densities, beetles cannot successfully attack healthy trees and must travel farther to find weakened hosts. Even so, the influence of beetle density on population spread is minor compared to the spatiotemporal variation in the overall (density-independent) scale of spread, with spatiotemporal heterogeneity accounting for 81% of the variance in our models. This spatiotemporal heterogeneity explains mountain pine beetle's erratic range expansion across western Alberta, which varied from 20 to 220 km annually.

KEYWORDS

bark beetle, density dependence, density-dependent dispersal, dispersal, dispersal kernel, long-distance dispersal, mountain pine beetle, range expansion, stratified dispersal

INTRODUCTION

Bark beetle outbreaks in North America are worsening due to climate change. Warmer winters promote larval survival (Carroll et al., 2006) and drought conditions weaken host trees' defenses (Alfaro et al., 2009; Creeden et al., 2014), both of which contribute to outbreak severity. Consequently, several species of bark beetles—most notably the mountain pine beetle (MPB; *Dendroctonus*

ponderosae Hopkins)—have had substantial socioeconomic impacts, harming timber-based industries and transforming the south-central region of British Columbia into a net carbon source (Abbott et al., 2009; Corbett et al., 2016; Dhar et al., 2016; Kurz et al., 2008).

Irruptive bark beetle species share similar natural histories (Raffa et al., 2015). These insects spend the majority of their life cycles beneath the bark of host trees. The larvae feed on the tree's phloem or cambium during

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warmer periods. Generation time varies from 2 years (i.e., semivoltine) to multiple generations per year (i.e., multivoltine), with overwintering beetles utilizing cryoprotectants to avoid freeze–thaw damage. Adults emerge briefly only in spring or summer for dispersal. Using aggregation pheromones and synchronized emergence (triggered by temperature-dependent developmental thresholds; Logan & Powell, 2001), the beetles launch coordinated attacks on trees (Raffa, 2001). Through these *mass attacks*, bark beetles can overwhelm a tree's defensive systems, which consist of both preexisting (constitutive) and stress-triggered (induced) compounds (Howe et al., 2020; Seybold et al., 2006). Once the tree's defenses are depleted, the beetles can successfully bore into their host (Raffa & Berryman, 1983).

To predict the spatial spread of irruptive bark beetles, we must understand their complex dispersal behavior, which is thought to be influenced by (among other things) wind, temperature, pheromones, host-tree volatiles, non-host-tree volatiles, microclimate, topography, and beetle body size (Amman & Logan, 1998; De La Giroday et al., 2011; Franklin & Grégoire, 1999; Gaylord et al., 2008; Jackson et al., 2008; Jones, 2019; Jones & Evenden, 2021; McCambridge, 1971; Safranyik et al., 1989; Shegelski et al., 2019; Wilson et al., 1996; Zhang et al., 1999). Despite extensive research, identifying the most important predictors of dispersal distance remains challenging. Even obtaining accurate in situ estimates of dispersal distance is difficult. Mark–recapture experiments have been used to estimate dispersal distance for MPB (Safranyik et al., 1989, 1992) and other bark beetle species (Dodds & Ross, 2002; Turchin & Thoeny, 1993; Werner & Holsten, 1997; Zolubas & Byers, 1995; Zurr, 1992), but studies using pheromone-baited traps may underestimate actual dispersal distances. A comparison of the only two MPB mark–recapture studies provides preliminary evidence for this bias: One study using pheromone traps found a median dispersal distance of less than 30 m (Safranyik et al., 1992), while passive traps in another study showed a median distance of 100 m (Safranyik et al., 1989, Table 2). Individual flight distances can be measured under laboratory conditions (e.g., Evenden et al., 2014), but these measurements may not accurately represent dispersal in complex natural environments.

An alternative approach, which we employ in this study, is to examine the spatial relationship between successive annual infestations as a proxy for beetle dispersal. This approach leverages remote-sensing data across a large spatiotemporal extent, allowing us to draw broad conclusions rather than those limited to specific years or locations. However, this method has key limitations: It assumes infestation density is proportional to beetle

density, ignores the impact of tree size on beetle production, overlooks changes in forest structure during outbreaks, and only captures successful beetle attacks. We address these constraints and their implications further in the *Discussion* section.

Understanding the dispersal of the MPB is particularly important, given that MPB has recently expanded its range from British Columbia into Alberta and now threatens boreal forests stretching eastward to Quebec (Bleiker, 2019; Cooke & Carroll, 2017; Safranyik et al., 2010). MPB is thought to exhibit three quasi-discrete dispersal behaviors, leading to a continuum of dispersal distances from zero meters to hundreds of kilometers. First, most beetles practice short-range dispersal, rarely venturing beyond 100 m from their natal trees (Robertson et al., 2009; Safranyik et al., 1992). Second, a minority of beetles, primarily females in good energetic condition characterized by their large body size and high lipid reserves (Shegelski et al., 2019), can disperse up to several kilometers under the canopy (Borden, 1993; Evenden et al., 2014; Robertson et al., 2007). This behavior is thought to be an evolutionarily stable bet-hedging strategy: Although most pioneers fail, the successful few that find high-quality host trees achieve high fitness (Jones et al., 2020; Kautz et al., 2016; Raffa, 2001). Third, a small fraction of beetles fly above the canopy, where atmospheric winds may carry them up to 300 km (Cerezke, 1989; Furniss & Furniss, 1972; Hiratsuka et al., 1982; Jackson et al., 2008). A mark–recapture study estimated that 2.5% of the MPB adults fly above the canopy (Safranyik et al., 1992). However, it is generally thought that MPB's propensity for this behavior could be much higher, especially in heavily infested stands (Bleiker, 2019; Bone & Altaweel, 2014; Carroll & Safranyik, 2003; De La Giroday et al., 2011).

The relationship between MPB population density and long-distance dispersal may also have management implications. During the so-called “hyperepidemic” of the 2000s in British Columbia, MPB jumped over the Rocky Mountains, a former geographic barrier, and spread to the central longitudes of Alberta in just four short years (Cooke & Carroll, 2017). Subsequent eastward spread was slow, which can be attributed to two main factors. First, eastern Alberta has lower pine volumes compared to the west (Bleiker, 2019; Safranyik et al., 2010). Second, Jack pine (*Pinus banksiana*), an evolutionarily novel host tree that inhabits eastern Alberta, may be less suitable for the beetle (Bleiker et al., 2023; Johnson et al., 2025; Srivastava & Carroll, 2023) compared to lodgepole pine (*Pinus contorta*), the predominant pine species in western Alberta and British Columbia. The primary risk of range expansion is now thought to be long-distance dispersal from high-volume pine stands in

central Alberta to high-volume stands in western Saskatchewan (Bleiker, 2019; Hodge et al., 2017). Therefore, a positive relationship between population density and long-distance dispersal would provide additional support for future suppression efforts targeted at high-density populations in central Alberta.

It is generally assumed that long-distance dispersal is more common at higher MPB densities, though there is little robust evidence, for example, mark-recapture experiments in stands with varying population densities, modeling studies that utilize observational data and allow for dynamic dispersal. Reid (2008) documented more medium and long-distance colonization events in monitoring zones (characterized by high beetle densities and no direct control measures) compared to management zones (with low beetle densities). It is unclear whether this pattern resulted from reduced mass attack success or from actual changes in dispersal behavior. On the other hand, Chen and Walton (2011) found a negative correlation between infestation densities and long-distance colonization events.

There are, however, several circumstantial reasons to expect a generally positive relationship between population density and dispersal distance at several different spatial scales. (1) The pheromone *exo-brevocomin* attracts female MPB at low concentrations but repels them at high concentrations (Rudinsky et al., 1974). This repellent effect drives beetles away from overcrowded areas, which may manifest as a larger-than-usual proportion of beetles flying above the canopy where wind may cause long-distance dispersal. Anti-aggregation pheromones like verbenone and frontalin (Borden et al., 1998; Ryker & Libbey, 1982) may also contribute to this dynamic. However, these pheromones trigger spillover attacks on trees within a couple of meters of their source (Bentz, 1996); this suggests these chemicals are effectual at short ranges, limiting any mediating effects on long-distance dispersal. (2) In the epidemic phase of an outbreak, there are sufficient beetles to attack and kill large pine trees. These trees have thick phloem, leading to beetles with a good energetic condition (Graf et al., 2012). Better energetic condition positively correlates with longer flight distances (Shegelski et al., 2019), which may result in a larger proportion of beetles adopting a medium-distance/pioneer dispersal strategy. (3) The two years in which many beetles dispersed from BC to Alberta (2006 and 2009; Carroll et al., 2017) occurred during the peak years of the outbreak in British Columbia. On the other hand, there are reasons to expect a negative relationship between MPB density and short/medium-distance dispersal. Perhaps beetles have to go farther at low beetle densities, since they do not have the numbers to attack

well-defended trees; instead, they must find recently compromised trees (Bleiker et al., 2014).

One mechanism thought to influence MPB dispersal is the availability of host trees. In theory, the absence of attractive semiochemicals—host-tree volatiles and aggregation pheromones, which are themselves derived from host-tree terpenoids—may prompt beetles to continue flying until suitable hosts are detected. Some modeling studies support this idea. Powell and Bentz (2014) developed a model implying that typical dispersal distances decrease (from kilometers to meters per day) as host tree density increases. Similarly, Goodsman et al. (2017) fit a statistical model showing lower beetle recruitment when the ratio of beetles to susceptible trees was high. In contrast, Reid (2008) found that variation in long-distance colonization events could not be explained by the cumulative area affected by MPB (inversely related to host-tree availability). Host-tree availability may affect dispersal dynamics, but we do not utilize this predictor for two practical reasons. Tree-size distribution data are rarely available, making it challenging to quantify suitable host-tree density. The low infestation densities in our Alberta study sites (compared to British Columbia; see citations in the *Discussion* section) mean that host-tree availability is dominated by initial host-tree density and is largely unaffected by beetle activity.

Here, we provide empirical evidence regarding the relationship between MPB density and both short- and long-distance dispersal using high-quality data and statistical models. Specifically, we utilized helicopter surveys, the most accurate remote-sensing method for bark beetle infestations (Coggins et al., 2008, Table 2). Our analysis employed simple models with a proven track record of characterizing MPB dispersal. While we anticipated a positive relationship, our findings revealed a small negative relationship. These results challenge existing assumptions and contribute to the ongoing dialogue about MPB dynamics and their implications for forest management.

METHODS

Data

Alberta's Ministry of Forest and Parks collects MPB infestation data annually through helicopter surveys (termed *Heli-GPS surveys*) and ground surveys. Around September, helicopters fly a "lawnmower" search pattern over large portions of Alberta, recording the locations of *red-topped trees*. These trees, characterized by rust-red needles, were first attacked the year prior. Because MPB is univoltine, the progeny from the red-topped trees have

already emerged and are likely infesting nearby *green-attack trees*, trees which have been successfully attacked but are not yet displaying red needles. Field crews are sent to the location of the red-topped tree to manually search for green-attack trees, using clues such as boring dust and pitch tubes. Infested trees are then “sanitized”, typically by cutting and burning. The quality assurance protocols ensure that at least 95% of the Heli-GPS points are within ± 30 m of the true location, and that 90% of the Heli-GPS points record the number of infested trees within a narrow error tolerance (the range changes depending on the true number of trees; Government of Alberta, 2016, p. 21). A previous study found that infestation-severity classifications were 92% accurate with Heli-GPS surveys (Nelson et al., 2006).

We focused on two patches in western Alberta, each approximately 2500 km² and separated by 50 km (edge-to-edge; Figure 1), deliberately chosen to minimize unmodeled environmental factors. Both patches are high-quality MPB habitats with high-biomass lodgepole pine forests, a moderate thermal regime, and relatively low topographical complexity. We focused on modeling the years 2010–2020, excluding earlier years due to the confounding influence of large beetle migrations from British Columbia in 2006 and 2009 (Carroll et al., 2017). These years cover the epidemic and declining phases of the outbreak in the study area (Appendix S1: Figure S2).

The raw Heli-GPS and ground survey data were rasterized into 30 × 30 m pixels, a resolution that roughly matches the Heli-GPS error margin. The number of infested trees in pixel x and in year t is computed as

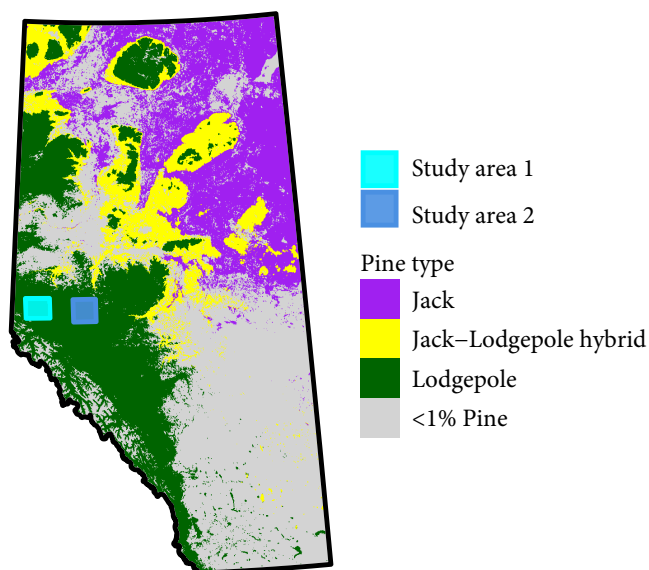


FIGURE 1 Location of the two study areas. Pine species data come from Cullingham et al. (2012). The gray pixels are areas where pines constitute less than 1% of total live aboveground biomass; data from Beaudoin et al. (2014).

$$I_t(x) = c_t(x) + r_{t+1}(x), \quad (1)$$

where $c_t(x)$ is the number of green-attack trees that were located and sanitized in the focal year, and $r_{t+1}(x)$ is the number of red-topped trees observed in the following year. While $I_t(x)$ is a suitable model output (i.e., a response variable), it is not a suitable model input—it contains controlled trees that do not produce any beetle progeny. Therefore, we define a modified input variable, $I_t^*(x) = I_t(x) - c_t(x)$.

Model outline

Our dispersal model considers all possible points of origin for each new infestation. Conceptually, the model operates by predicting the locations of infestations in the focal year (*offspring infestations*) stemming from each infestation in the prior year (*parental infestations*). Then, these individual-level distributions are normalized to create a spatially explicit probability mass function (Figure 2).

Specifically, we utilize a radially symmetric Student's t distribution, a dispersal kernel that has been validated by previous research (Johnson et al., 2026). Due to radial symmetry, the kernel can be parameterized as a function of the Euclidean distance between coordinates, $r = \text{dist}(x, y)$, where x and y index pixels in 2D space.

While we use “dispersal kernel” here for its familiarity, this terminology requires clarification. Rather than directly tracking beetle movement, we are modeling the spatial relationships between successive years of infestations. For this reason, we specifically use “colonization distance” in the *Results* and *Discussion* sections to help distinguish between our observable measure (spatial redistribution of infestations) and the unobservable phenomenon of interest (actual beetle dispersal).

The Student's t dispersal kernel is written as

$$D(r) = \frac{(\nu - 1) \left(\frac{r^2}{\nu \rho^2} + 1 \right)^{\frac{1}{2}(\nu - 1)}}{2\pi\nu\rho^2}. \quad (2)$$

The likelihood function is computed by convolving parental infestations with the dispersal kernel:

$$B_t(y) = \sum_x I_{t-1}^*(x) D(\text{dist}(y, x)) (\Delta x)^2, \quad (3)$$

with $\Delta x = 0.03$ km as the spatial resolution. The result is *beetle potential* B_t , which represents the mean number of offspring infestations received by each pixel; here, “mean” refers to an average of instantiations of a stochastic dispersal process.

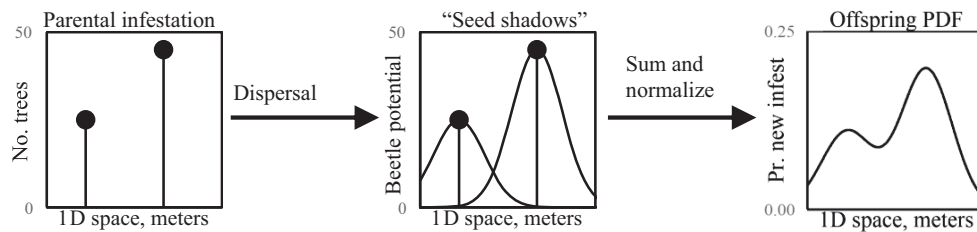


FIGURE 2 Graphical representation of our colonization models in one dimension. A dispersal model is applied to each parental infestation, creating surfaces that represent the potential of offspring infestations stemming from parental infestation, termed “seed shadows” in the plant-focused dispersal literature. These are summed and normalized to produce a likelihood function for new infestations. Figure originally appears in Johnson et al. (2026).

The beetle potential is rescaled to produce a likelihood surface:

$$\pi_t(x) = \frac{B_t(x)}{\sum_y I_{t-1}^*(y)} \quad (4)$$

The convolution is performed for all infestations within the approximately 50×50 km study area, but the likelihood is only calculated using the offspring infestations in a smaller inset of the study area, defined by a negative 10-km buffer. Without the buffer, the model might incorrectly infer that infestations near the edge of the study area were seeded from far-away infestations, simply because the model cannot “see” infestations just outside of the study area.

Offspring infestations are treated as independently and identically distributed events, and thus, the log-likelihood in year t is

$$\mathcal{L}_t = \sum_{x: I_t(x) > 0} I_t(x) \times \log(\pi_t(x)). \quad (5)$$

Density-dependent dispersal models

We examine the relationship between infestation density and colonization distance in two ways. First, we performed a *nonlocal dispersal analysis*, fitting the dispersal kernel parameters separately to each of the 22 unique combinations of year and study area. Then, we examined the relationship between infestation density and summary statistics of the fitted dispersal kernel (e.g., the median colonization distance). This approach has the benefit of not assuming the functional form of the relationship between infestation density and colonization distance; however, it implicitly assumes that the infestation density, averaged over the entire study area, is representative of the infestation density that beetles experience in their local neighborhoods.

Individual beetles likely have a sensory neighborhood with a radius of 20–1000 m. Evidence from small-scale spatial maps and mechanistic pheromone diffusion models suggests that beetles primarily respond to aggregation pheromones that are a maximum of 20–100 m from their tree of origin (Mitchell & Preisler, 1991, figs. 4 and 5; Strohm et al., 2013). However, some beetles are caught in pheromone traps up to 1 km from a point of release, likely due to a combination of diffusive flight and pheromone detection (Barclay et al., 1998).

To represent MPB’s limited sensory abilities, we also utilized *neighborhood-based dispersal models*. These are hierarchical models in which the dispersal parameters (ρ and ν) are location-specific and depend on the infestation density in a local neighborhood:

$$\rho_t(x) = \exp \left[\alpha_0 + \alpha_1 \log \left(\sum_{y \in N(x)} I_{t-1}^*(y) \right) \right], \quad (6)$$

$$\nu_t(x) = \exp \left[\beta_0 + \beta_1 \log \left(\sum_{y \in N(x)} I_{t-1}^*(y) \right) \right] + 1. \quad (7)$$

Here, $N(x)$ is a circular neighborhood around the location x . We fit the hierarchical models separately for several fixed neighborhood sizes: 0 (i.e., only density in the focal raster cell is used), 0.003, 0.025, 0.636, 2.545, and 10.179 km². These neighborhoods led to similar model performance (Appendix S1: Figure S1). Consequently, all further analysis was conducted with the 0.025-km² neighborhood, which is equivalent to a sensory radius of 90 m.

The structure of the neighborhood-based models is justified by graphical evidence: the log-log linear relationship between the dispersal parameters (ρ and ν) and infestation density emerges from the nonlocal dispersal analysis (Appendix S1: Figure S3). We fit the neighborhood model to each year $\times$ site subset of the data to avoid spurious correlations between colonization distance and

infestation density, which may be driven by third variables that vary substantially from year to year, for example, minimum winter temperature, wind speed during the flight period, and the density of suitable host trees remaining.

All models were fit with maximum likelihood estimation using the Nelder–Mead optimization algorithm. To monitor for local optima, we repeatedly fit each data subset with random initial parameters and manually checked that the maximum likelihood parameters converged to nearly identical values. All analyses were performed in R (R Core Team, 2020).

In addition to the primary neighborhood-based model, we tested an expanded model to explore potential hump-shaped or U-shaped relationships between infestation density and colonization distance. This expanded model modifies Equations (6) and (7) to include quadratic terms (details in Appendix S2).

We faced significant computational challenges due to the necessity of brute force convolution, which scales as $O(n^4)$, where n is the linear dimension of our study area. Parameter estimates were unstable with larger pixel sizes, as most infestations occurred within 100 m of the natal tree. To reduce the computational burden, we limited dispersal kernels to a 5-km radius from parental trees. Inferences made about dispersal across longer distances are therefore based on extrapolating the shape of the dispersal kernel. Offspring infestations that appeared further than 5 km from any parental infestations were extremely rare; to avoid an infinite log-likelihood stemming from these few observations, we added machine epsilon to each location's probability mass and renormalized. Despite these adjustments, computations still required approximately 48 h on a 40-core computer.

To quantify the relative importance of density dependence versus spatiotemporal heterogeneity, we performed variance partitioning on the logarithm of both the median and 95th percentile colonization distances. Using the law of total variance, the variation is split into two components: variation across infestation densities within areas and years, and variation between areas and years. This approach allows us to determine whether differences in dispersal are primarily driven by local beetle density or by unobserved environmental factors that vary across time and space.

RESULTS

The nonlocal dispersal analysis—where dispersal kernels are fit to 22 year $\times$ site subsets of the data—reveals a negative association between infestation density and colonization distance (Figure 3). This relationship applies to

both short- and long-distance colonization, respectively, characterized by the median and 95th percentile of predicted colonization distances. A simple linear model, treating each fitted dispersal kernel as an observation, predicts that long-distance colonization decreases from 10 to 2 km across the data range. However, this negative relationship is noisy ($R^2 = 0.08, 0.19$ for the median and 95th percentile, respectively), even without accounting for conditional non-independence, that is, the fact that successive years have similarly shaped colonization kernels.

The neighborhood-based dispersal models—where dispersal parameters depend on the density of nearby infestations—also show an overall negative relationship between infestation density and colonization distance (Figure 4). The quadratic neighborhood model, which tests for a potential U-shaped relationship between infestation density and colonization distance, produced results qualitatively similar to the simpler linear model (Appendix S2). However, the predominant pattern is one of spatiotemporal heterogeneity. There are substantial differences between years and study areas regarding the overall scale of dispersal. To a lesser extent, there is heterogeneity in the sign of the relationship between density and dispersal.

To cut through this heterogeneity, we can compute the median of the dispersal parameters (ν and ρ) across years and study areas for a range of neighborhood infestation densities. This provides a typical relationship between density and dispersal, which is small and negative (Figure 5). To be more quantitative, we may consider how dispersal changes as neighborhood infestation density increases from the 10th to the 95th percentile (1.1–8.4 infestations/ha). The median colonization distance decreases from 58 to 42 m, whereas the 95th percentile of colonization distance decreases from 3.4 to 1.6 km (Table 1). The difference in colonization distance between low and high infestation densities increases (both in relative and absolute terms) for beetles that travel further from their origin. In other words, the effect of infestation density is more pronounced in the tails of the dispersal kernel.

Our variance partitioning analysis revealed that most of the variation in colonization distance is attributable to spatiotemporal heterogeneity rather than infestation density. For the median colonization distance, 82% of the total variance in the logarithm of distance is explained by differences between areas and years, compared to only 18% explained by variation across infestation densities within areas and years. Similarly, for the 95th percentile of colonization distance, nearly three-quarters of the variance is explained by spatiotemporal heterogeneity (Table 2).

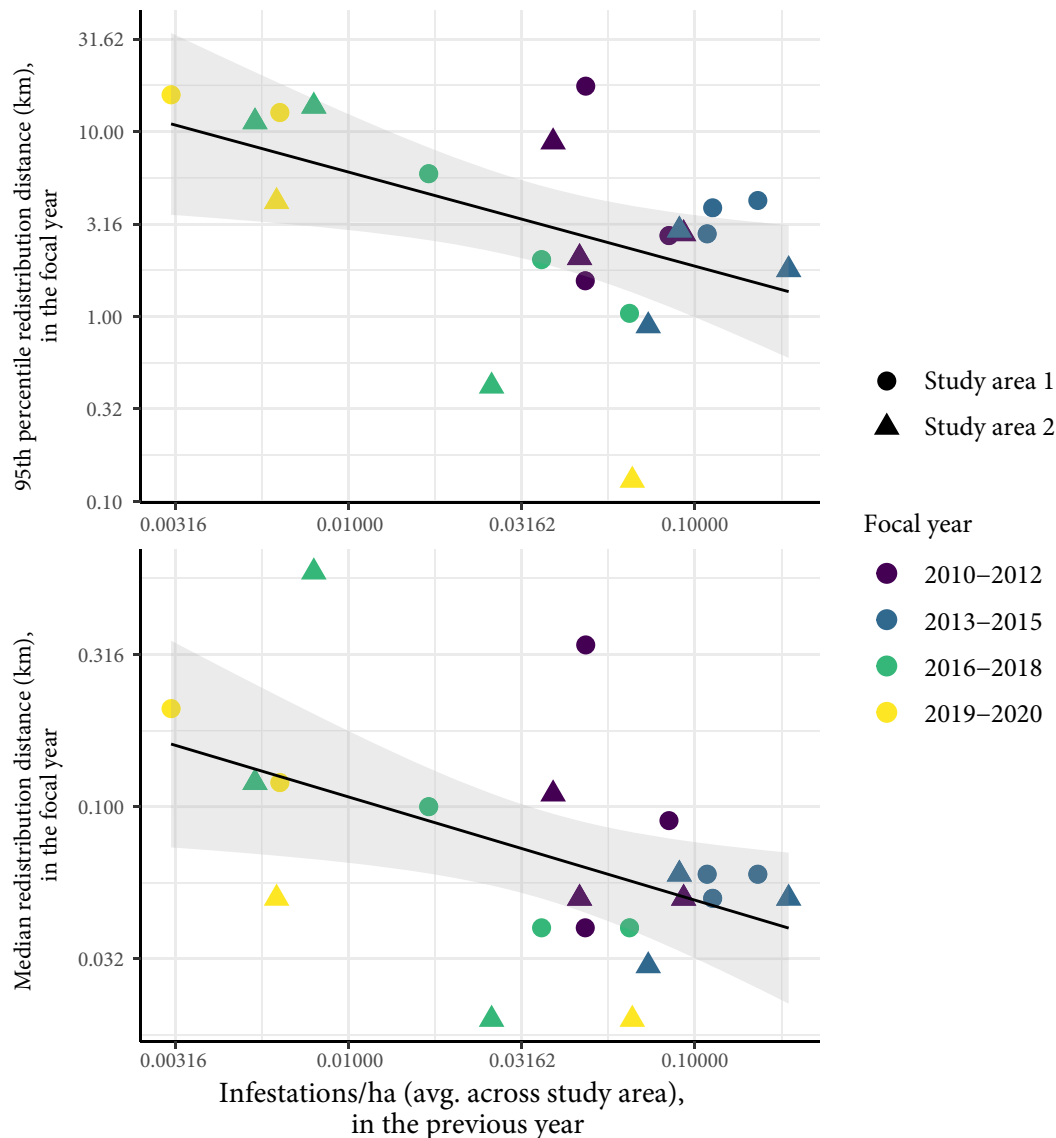


FIGURE 3 There is a negative association between global infestation density (averaged across a study area) and colonization distance. Each point represents a unique combination of study year and study area.

DISCUSSION

We observe a weak negative association between infestation densities and infestation spread, suggesting that MPB exhibits negative density-dependent dispersal. While statistical models can be blunt instruments and are subject to various limitations (which we discuss extensively below), there are several reasons to believe that the negative density dependence is real and not a statistical artifact. The biological mechanism is plausible: Higher population densities enable beetles to mass attack nearby healthy trees instead of seeking out weakened ones. Anecdotal field observations support this pattern, with only clusters of 3–6 (or more) infested trees likely to produce nearby infestations in the following year (Katherine Bleiker, personal communication).

Interestingly, the estimated effect of infestation density on colonization distance is dwarfed by spatial and temporal heterogeneity in the typical (density-independent) level of colonization distances. Visually, the shape of the dispersal kernel changes very little as infestation density changes (Figure 5). Quantitatively, we found that spatiotemporal heterogeneity accounted for around 80% of the total variance in log-transformed colonization distance. This heterogeneity is evident in the erratic eastward expansion of MPB across Alberta, varying from 20 to 220 km per year (Cooke & Carroll, 2017). There is no shortage of factors that could plausibly explain this variability: the occurrence of wind storms during the beetles' 1- to 4-week flight period; adverse weather conditions during the flight period (e.g., too much wind or precipitation, Gray et al., 1972; Musso, 2023, Ch. 2); cold winter

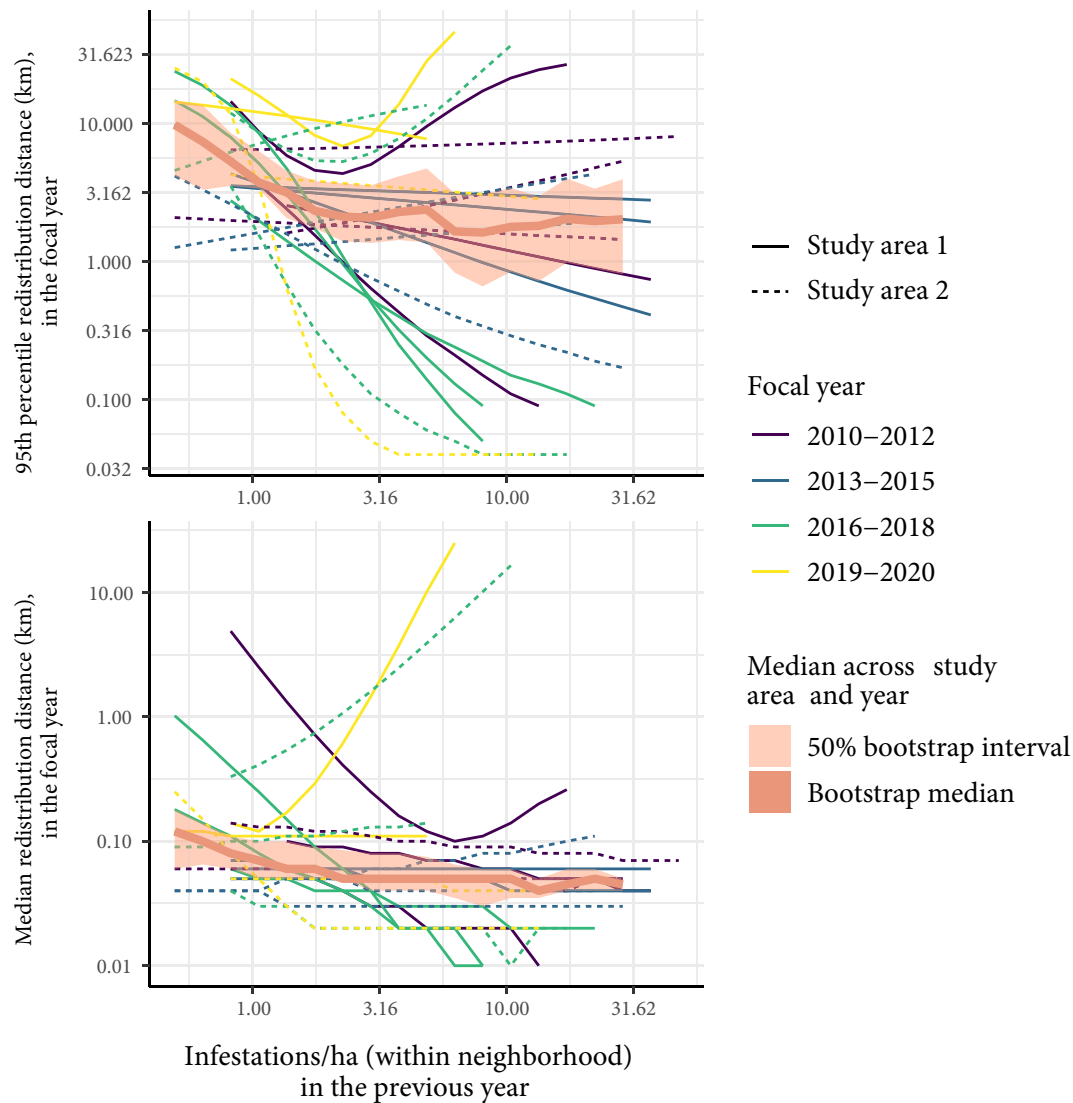


FIGURE 4 The neighborhood-based dispersal model predicts a negative relationship between infestation density and colonization distance, but the sign and strength of the relationship can vary substantially across space and time. The bootstrap intervals show the uncertainty in the spatiotemporal median of colonization distance.

temperatures affecting development and subsequent flight capability (Nijholt, 1967; Shogelski et al., 2019); tree defensive compounds—thought to vary over large spatial scales (Cudmore, 2009; Clark et al., 2012)—affecting development and subsequent flight capability (Manning & Reid, 2013); and volatiles from deciduous nonhost trees affecting flight propensity (Jones & Evenden, 2021).

Previous research on density-dependent dispersal provides a framework for interpreting our findings on MPB dispersal patterns. The literature shows that positive density-dependent dispersal is favored under specific conditions: when high-quality patch locations shift over time (McPeck & Holt, 1992; Rodrigues & Johnstone, 2014), and/or when dispersal costs are minimal (Poethke & Hovestadt, 2002; Travis et al., 1999). Positive density dependence is selected against when local growth

dynamics exhibit positive density dependence (Goodsman & Lewis, 2016; Rousset, 2012; Travis et al., 1999). MPB experiences a *strong Allee effect*, an extreme form of positive density dependence where per capita growth rates become negative at sufficiently low population densities: Negative growth occurs because there are not enough beetles to successfully mass attack a single tree (Boone et al., 2011; Goodsman et al., 2017; Raffa & Berryman, 1983; Safranyik et al., 1975). MPB dispersal is exceptionally costly, with 40%–80% of emerging beetles dying during the dispersal/host-finding phase of the life cycle (Safranyik, 1989; Latty, 2007; similar mortality has been observed in the southern pine beetle, Pope et al., 1980). This high mortality rate stems from several factors. MPB have limited survival time outside their host trees (Fuchs & Borden, 1985; Reid & Reid, 2008), and

prolonged exposure increases the chance of avian predation (Rust, 1930; Stallcup et al., 1963). Further, beetles covering greater distances are less likely to engage in pioneer attack behavior, likely due to depleted energy reserves (Jones, 2019; Jones et al., 2020). These features

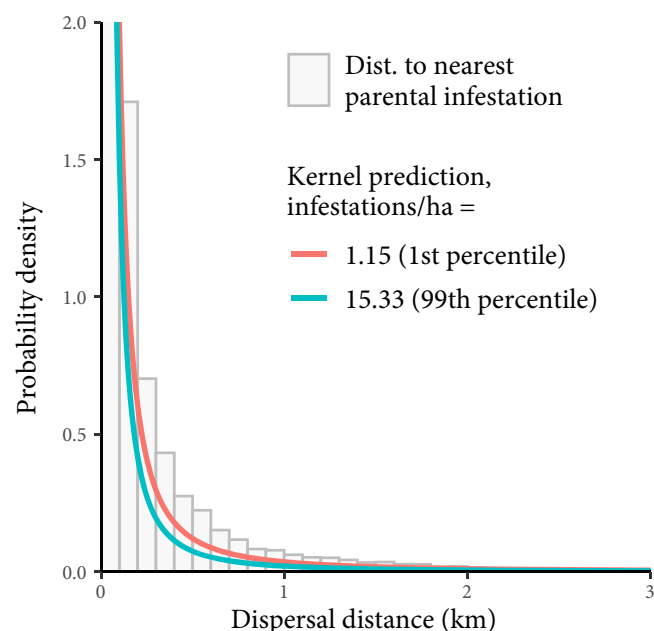


FIGURE 5 The estimated probability of colonization as a function of distance, that is, “dispersal kernels.” The two curves represent dispersal patterns under low and high infestation densities, demonstrating that low infestation densities are associated with slightly greater colonization distances. The plotted curves show marginal dispersal kernels: $\int_0^{2\pi} D(r) r d\theta = 2\pi r D(r)$, where r is the distance in kilometers, θ is the angle in radians, and $D(r)$ is the full density (Equation 2). The histogram gives the frequency of distances between infestations and the closest “parental infestation”, that is, infestations in the previous year.

of MPB biology—a strong Allee effect and high dispersal costs—form the selective basis for near-zero or even negative density-dependent dispersal.

The theoretical correlates for the evolution of negative density-dependent dispersal—costly movement and strong Allee effects—are prevalent among forest pest insects, suggesting this behavioral response may be a general adaptation in forest pest populations. These insects face significant energy costs during active dispersal, while passive wind dispersal risks landing in unsuitable habitats (e.g., Cheptou et al., 2008). Allee effects are widespread among forest pests, particularly in bark beetles that employ aggregation pheromones to coordinate mass attacks on host trees (Weed et al., 2015); this category includes species such as the North American spruce beetle (*Dendroctonus rufipennis* Kirby; Massey & Wygant, 1954; Anderson et al., 2010) and the western pine beetle (*Dendroctonus brevicomis*; Homicz et al., 2022). Evidence of Allee effects has also been documented in forest Lepidoptera like gypsy moths (Johnson et al., 2006)

TABLE 2 Heterogeneity across study areas and years accounts for most of the variation in colonization distance: 81% of the variation in the logarithm of the median colonization distance, and 73% of the variation in the logarithm of the 95th percentile of colonization distance.

Source of variation	Variation in ...	
	Log (median dist.)	Log (95th perc. Dist.)
Across infestation densities, within areas and years	0.15	0.37
Between areas and years	0.66	1.00

TABLE 1 Relationship between infestation density and typical colonization distance.

Infestations/ha		Median colonization distance			95th percentile colonization distance		
Quantile	Value	Median	BI _{0.025%}	BI _{97.5%}	Median	BI _{0.025%}	BI _{97.5%}
0.100	1.149	0.058	0.036	1.029	3.398	1.465	9.703
0.250	1.533	0.055	0.031	0.522	2.648	1.177	8.504
0.500	1.916	0.054	0.030	0.442	2.080	0.804	7.463
0.750	3.448	0.051	0.024	1.332	1.967	0.307	8.569
0.900	6.130	0.044	0.015	5.982	1.619	0.064	15.309
0.950	8.429	0.042	0.012	5.487	1.572	0.043	15.937
0.975	11.111	0.043	0.015	9.249	1.686	0.043	20.600
0.990	15.326	0.041	0.015	0.078	1.766	0.068	12.235

Note: Quantiles are calculated across all neighborhoods, years, and study areas. Distance numbers represent the median across years and study areas where the given infestation density was observed (to avoid extrapolating beyond the data’s range). Bootstrap intervals (BI) for the median were calculated with the same procedure. Empirical quantiles for probabilities 0.01, 0.025, and 0.05 were identical to the 0.1 probability (1.149 infestations/ha) and were thus omitted.

and spruce budworm (Régnière et al., 2013; but see Royama, 1984).

Our finding of a negative relationship between observed variables (infestation density and colonization distance) provides evidence for density dependence in the unobserved variables (actual beetle density and dispersal distance), partly because our methodology creates detection bias toward a positive relationship in observed variables when no true density-dependent dispersal exists. This methodological bias stems from two factors: MPB's well-documented Allee effect (requiring sufficient numbers for successful tree attacks) and our use of successive-year infestation patterns as proxies for actual beetle movement. Long-distance dispersers rarely establish detectable infestations because too few beetles reach distant locations to overcome tree defenses; conversely, at high densities, more beetles disperse, increasing the proportion of successful distant colonization events. The fact that we instead observed fewer long-distance events at higher densities—directly contrary to this methodological bias—reinforces the hypothesis of negative density-dependent dispersal.

Although our study does not support a positive relationship between beetle density and long-distance dispersal, this possibility cannot be definitively ruled out. Analysis of a dispersal model with quadratic effects, which could reveal a U-shaped relationship between infestation density and colonization distance, suggests this dynamic is not occurring in Alberta (Appendix S2). However, a positive relationship might exist but only activate at higher beetle densities than those observed—average infestation densities in Alberta are 2–3 orders of magnitude lower than in British Columbia. In British Columbia, fixed-wing aerial surveys (Westfall & Ebata, 2007) identified large areas with moderate (15%–30% canopy damage) to severe (over 30% canopy damage) infestation levels. Given that lodgepole pine grows in dense stands exceeding 1000 trees per hectare (Meddens et al., 2011; Nigh et al., 2008; Pfeifer et al., 2011), these damage percentages translate to tens to hundreds of infested trees per hectare during peak outbreak years. This contrasts sharply with helicopter survey data from Alberta, which showed infestation densities of fewer than 0.1 trees killed per hectare across most of the 2010–2020 period (fig. 3 in Johnson & Lewis, 2025). Unfortunately, Heli-GPS data for British Columbia during the peak outbreak years were never collected. An alternative test for density-dependent dispersal involves examining invasion speed, as positive density-dependent dispersal leads to acceleration of the invasion front over time (Dwyer & Morris, 2006; Travis et al., 2009). This analysis would be uninformative for two reasons: a limited sample size stemming from the short duration of

MPB's range expansion across western Alberta (approximately 2005–2009) and the fact that acceleration in invasion speed could alternatively be attributed to MPB's Allee effect (Kot et al., 1996; Wang et al., 2002).

Our analysis has important limitations stemming from several unmodeled factors, particularly tree size. Larger trees both produce more beetle offspring (Cudmore, 2009; Reid, 1963) and are preferentially selected by beetles for attack (Klein et al., 1978). The early part of our study period contained a higher proportion of large trees, putatively resulting in larger beetles (which disperse farther; Graf et al., 2012; Shegelski et al., 2019) and more beetles produced per tree. By failing to account for tree size, our model will tend to overestimate typical/tree-size-independent dispersal distances (i.e., the expected dispersal distance in a stand of uniformly median-sized trees) and underestimate beetle population sizes in those early years. Both effects artificially inflate the slope of infestation density versus colonization distance, that is, a positive bias. On the other hand, the abundance of suitable nearby trees may have shortened dispersal distances, potentially causing our model to underestimate typical dispersal distances. But this negative bias may be offset by behavioral plasticity as beetles prefer smaller trees during population declines (Boone et al., 2011; Howe et al., 2022). Given all these opposing effects, the net influence of tree size and suitable tree density remains uncertain.

Spatial heterogeneity in host-tree density presents another source of potential bias. Areas with high infestation densities typically correlate with high host-tree abundance. Due to regression to the mean, distant locations are statistically more likely to have lower host-tree densities than these focal high-infestation areas. Consequently, beetles dispersing longer distances from high-density areas may fail to find suitable hosts, resulting in unobserved dispersal events. This phenomenon would create an apparent bias toward shorter colonization distances in high-infestation locations. However, we believe this heterogeneity has minimal impact on our findings. We specifically chose study areas with consistently high-biomass lodgepole pine forests to minimize heterogeneity. Also, MPB frequently appears in stands of nonhost trees and can move through clear-cuts and thinned areas without significant population losses (Carroll et al., 2020; Reid, 2008).

A consequence of the aforementioned limitations is that our results may not generalize across all phases of a MPB outbreak. Researchers have traditionally delineated three distinct phases: incipient, epidemic, and declining phases (Safranyik & Carroll, 2006). Our study primarily captures the epidemic and declining phases, when large host trees become increasingly scarce and beetles may

exhibit altered host selection behavior (Boone et al., 2011). Although winter temperatures are known to affect beetle mortality (Safranyik & Linton, 1991; Stahl et al., 2006), using temperature data to adjust infestation densities lies beyond the scope of this study. However, our colonization distance estimates appeared unaffected by the colder-than-usual winter of 2018–2019, and the particularly severe winter of 2019–2020 (MacQuarrie et al., 2019), as shown in residual plots (Appendix S1: Figure S4).

Finally, our statistical methodology introduces two limitations. First, our dispersal kernels are parameterized using infestation densities within a 5-km radius (see *Methods*) due to computational constraints, meaning that inferences about dispersal beyond 5 km rely on extrapolation. Second, our models assume that the location of every infestation relative to the source is an independently distributed random variable given by the dispersal kernel, whereas in reality, MPB infestations are more spatially autocorrelated than expected (after accounting for local dispersal), due to aggregation behavior and non-isometric dispersal patterns. Consequently, the standard errors for the dispersal hyperparameters ($\alpha_0, \alpha_1, \alpha_2, \beta_0, \beta_1$, and β_2) are smaller than they should be. The standard solution, a spatial Gaussian process, is computationally infeasible with our large dataset. Therefore, instead of reporting and emphasizing the standard errors of dispersal hyperparameters, we acknowledge uncertainty by focusing on between-year and between-site heterogeneity (e.g., Figure 4).

Despite these limitations, the weight of evidence from our analysis supports a small negative relationship between MPB density and dispersal distance, at least during the epidemic and declining phases of the Alberta outbreak. This conclusion is strengthened by agreement across both nonlocal- and local-neighborhood models, which examine temporal and spatial variation in infestation densities respectively. The negative relationship is consistent with a biologically plausible mechanism—mass attacks on nearby trees are more likely to be successful at high beetle densities—and is further strengthened by the expectation of a positive bias due to our reliance on observed infestations as a proxy for beetle presence and dispersal. The negative association aligns with MPB's well-documented Allee effect and the substantial costs of dispersal. However, given the various sources of potential bias and the relatively low infestation densities observed in Alberta, we cannot make definitive claims. It remains possible that positive density-dependent dispersal occurs at higher densities, such as those recorded in British Columbia circa 2005–2010.

While our results challenge the prevailing notion of positive density dependence in MPB dispersal, they also highlight a more striking pattern: the effect of infestation

density on colonization distance is overshadowed by spatiotemporal heterogeneity. Future research on MPB dispersal should prioritize explaining this variation. We suggest starting by investigating how wind speed and direction during the MPB's flight period affect dispersal, given that windstorms have clearly facilitated MPB's journey across the Canadian Rocky Mountains (Chen & Jackson, 2017; Jackson et al., 2008), and that wind direction is highly predictive of spruce budworm dispersal (Sturtevant et al., 2013). Additional environmental factors—including adverse weather conditions, winter temperatures, tree defensive compounds, forest structure, and deciduous tree volatiles—warrant further investigation. Understanding the drivers of dispersal heterogeneity, rather than density-dependent effects alone, may ultimately prove more valuable for predicting and managing future MPB outbreaks.

AUTHOR CONTRIBUTIONS

Evan C. Johnson conceived the project, performed the analysis, and wrote the first draft; both authors contributed critically to the drafts and gave final approval for publication.

ACKNOWLEDGMENTS

The authors would like to thank K'evan Rastello, Micah Brush, and Leanne Petro for their feedback and relevant conversations. Funding for this research has been provided through grants to the TRIA-For Project to Mark A. Lewis from Genome Canada (Project Number 18202) and the Government of Alberta through Genome Alberta (Grant Number L20TF). Mark A. Lewis is grateful for funding from the NSERC Discovery Grant program (RGPIN-2024-04792).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Code and the publicly available data (Johnson, 2025) are available from Figshare: <https://doi.org/10.6084/m9.figshare.30646574>. Heli-GPS data may be requested from the Alberta Ministry of Forestry and Parks (Mike Undershultz, Senior Forest Entomologist, Mike.Undershultz@gov.ab.ca).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Johnson, Evan C., and Mark A. Lewis. 2026. "Evidence of Negative Density-Dependent Dispersal in an Invasive Forest Pest." *Ecosphere* 17(4): e70544. <https://doi.org/10.1002/ecs2.70544>